

Mathematics of Data Science - CA 3

Name: _____

1. Recall the n -dimensional sphere centered at the origin

$$\mathcal{S} := \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x}\| = 1\},$$

and the n -dimensional cube centered at the origin

$$\mathcal{C} := \left[-\frac{1}{2}, \frac{1}{2}\right]^n.$$

Does the 1000-dimensional unit sphere centered at the origin contain the corners of the 1000-dimensional cube centered at the origin? Explain your answer.

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3 pts ☐

2. Let $\mathbf{A} \in \mathbb{R}^{1000 \times 1000}$ be a random matrix with independent and identically distributed (i.i.d.) entries $\mathbf{A}_{ij} \in \mathbb{R}$ such that

$$\mathbb{E}[\mathbf{A}_{ij}] = -1, \quad \text{Var}(\mathbf{A}_{ij}) = 9.$$

Let $\mathbf{U}$ be the all-ones matrix (i.e., $\mathbf{U}_{ij} = 1$ for all i, j).

a) Compute the mean and variance of the *entries* of $\mathbf{W} = \mathbf{A} + \mathbf{U}$, i.e., $\mathbb{E}[\mathbf{W}_{ij}]$ and $\text{Var}(\mathbf{W}_{ij})$.

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2 pts ☐

b) For a unit vector $\mathbf{v}$ (i.e., $\|\mathbf{v}\|_2 = 1$), compute $\mathbb{E}[(\mathbf{W}\mathbf{v})_i^2]$, i.e., the expectation of the i -th coordinate of $\mathbf{W}\mathbf{v}$ to the power of 2.

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2 pts

c) Compute $\mathbb{E}[\|\mathbf{W}\mathbf{v}\|_2^2]$.

1 pts

d) Let $\mathbf{W}$ be a realization of $\mathbf{W}$. Assume there exists a unit vector $\mathbf{v}^*$ such that $\|\mathbf{W}\mathbf{v}^*\|_2^2 = 10^4$, what can we say about the operator norm $\|\mathbf{W}\|_2$?

2 pts

1. No. Consider the distance between the origin $o = (0, \dots, 0)$ and a corner $v = (1/2, \dots, 1/2)$ of the cube:

$$\|v - o\|_2^2 = \sum_{j=1}^{1000} \left(\frac{1}{2}\right)^2 = 1000 \cdot \frac{1}{4} = 250.$$

Thus

$$\|v - o\|_2 = \sqrt{250} > 1,$$

so this corner lies outside the unit sphere of radius 1.

Moreover, every corner of the cube has coordinates $\pm 1/2$ in each component, so all corners have the same distance $\sqrt{250}$ from the origin. Therefore, none of the corners of the cube are contained in the unit sphere.

2. a)

$$\mathbb{E}[\mathbf{W}_{ij}] = \mathbb{E}[\mathbf{A}_{ij} + \mathbf{U}_{ij}] = \mathbb{E}[\mathbf{A}_{ij}] + \mathbb{E}[\mathbf{U}_{ij}] = -1 + 1 = 0,$$

and

$$\text{Var}[\mathbf{W}_{ij}] = \text{Var}[\mathbf{A}_{ij} + \mathbf{U}_{ij}] = \text{Var}[\mathbf{A}_{ij}] = 9.$$

b) We have

$$\mathbb{E}[(\mathbf{W}\mathbf{v})_i^2] = \mathbb{E}\left[\left(\sum_{j=1}^{1000} \mathbf{W}_{ij} \mathbf{v}_j\right)^2\right] = \mathbb{E}\left[\sum_{j,k=1}^{1000} \mathbf{W}_{ij} \mathbf{W}_{ik} \mathbf{v}_j \mathbf{v}_k\right] = \sum_{j,k=1}^{1000} \mathbb{E}[\mathbf{W}_{ij} \mathbf{W}_{ik}] \mathbf{v}_j \mathbf{v}_k.$$

Since $\mathbb{E}[\mathbf{W}_{ij} \mathbf{W}_{ik}] = 9 \cdot \delta_{jk}$ because entries are i.i.d. and $\|\mathbf{v}\|_2^2 = 1$, we have

$$\mathbb{E}[(\mathbf{W}\mathbf{v})_i^2] = 9 \sum_{j=1}^{1000} \mathbf{v}_j^2 = 9.$$

$$\text{c) } \mathbb{E}[\|\mathbf{W}\mathbf{v}\|_2^2] = \sum_{i=1}^{1000} \mathbb{E}[(\mathbf{W}\mathbf{v})_i^2] = \sum_{j=1}^{1000} 9 = 9000.$$

$$\text{d) } \|\mathbf{W}\|_2 = \max_{\mathbf{v}: \|\mathbf{v}\|_2=1} \|\mathbf{W}\mathbf{v}\|_2 \geq \|\mathbf{W}\mathbf{v}^*\|_2 = \sqrt{10^4} = 10^2.$$